

APEX Developer Quickstart (Repository-Accurate)

Root: `backend/` — application package under `backend/src` (import path `src` on `PYTHONPATH`).

Run the API Locally

- 1. `cd backend && poetry install`
 - 2. `cp .env.example .env` — fill `database_url`, `supabase_jwt_secret`, `openai_api_key`, `gemini_api_key`, etc. (`config/settings.py` lists fields).
 - 3. `poetry run alembic upgrade head`
 - 4. Set `PYTHONPATH` to `backend/src` (PowerShell: `$env:PYTHONPATH = "$PWD\src"` from `backend/`).
 - 5. `poetry run uvicorn src.app:app --reload --host 0.0.0.0 --port 8000`
- Health: `GET /health`
 - OpenAPI: `http://localhost:8000/v1/docs`
 - Media (local storage): `/media` mount (`app.py`)
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Plugin Code Locations

Item	Path
Load list	<code>src/plugins/load_plugins.py</code> — <code>PLUGIN_MODULES</code>
Registry	<code>src/plugins/registry.py</code> — <code>PluginRegistry</code>
Dynamic import / cached getters	<code>src/core/plugin_manager.py</code> — <code>_load_class_from_path</code> , <code>get_patient_model</code> , <code>get_evaluator</code> , <code>get_metrics_plugins</code>
Startup hook	<code>src/core/events.py</code> — <code>load_plugins()</code>

Add a Plugin (Minimal Steps)

- 1. Create a module under `src/plugins/...` with a class implementing the matching `Protocol` in `src/interfaces/`.
- 2. Set `name: str = "full.module.path:ClassName"` (must match how you reference it in `settings/case/session`).
- 3. On module import, call `PluginRegistry.register_patient_model(name, Cls)`, `register_evaluator`, or `register_metrics_plugin`.
- 4. Append the **module path** (dotted, no class) to `PLUGIN_MODULES` in `load_plugins.py`.
- 5. Optionally set defaults in `.env` / settings: `patient_model_plugin`, `evaluator_plugin`, `metrics_plugins` (list).

Reference implementations: `default_llm_patient.py`, `apex_hybrid_evaluator.py`, `apex_metrics.py`.

Configure Which Plugin Runs

Layer	Mechanism
Defaults	<code>Settings</code> in <code>config/settings.py</code> — e.g. default evaluator <code>plugins.evaluators.apex_hybrid_evaluator:ApexHybridEvaluator</code>
Case	DB columns on <code>cases</code> : <code>evaluator_plugin</code> , <code>patient_model_plugin</code> , <code>metrics_plugins</code> (JSON array text)
Single session	<code>POST /v1/sessions</code> body: <code>evaluator_plugin</code> only (optional override); patient/metrics come from case/settings

Resolved values are **frozen** on `sessions` columns when the row is created (`SessionService.create_session`).

Tests

- `backend/tests/plugins/` — registry and plugin behavior
- Clear `lru_cache` on `plugin_manager.get_*` if tests change `settings`

Scoring and Metrics (Important)

- Trainee flow triggers `ScoringService.generate_feedback` via `POST /v1/sessions/{id}:close` (`sessions_controller.close_session_and_get_feedback`).
- `generate_feedback` calls `evaluator.evaluate`, then `_run_and_store_metrics_plugins`.
- Evaluator plugins call `generate_feedback_rule_only`, `generate_feedback_hybrid`, or `generate_feedback_hybrid_v2` — those methods **do not** run metrics plugins.

Useful API Examples (prefix `/v1`)

Action	Method	Path
Create session	POST	<code>/sessions</code>
Text turn	POST	<code>/sessions/{id}/turns</code>
Close + feedback	POST	<code>/sessions/{id}:close</code>
Get feedback	GET	<code>/sessions/{id}/feedback</code>
Admin plugin list	GET	<code>/admin/plugin-registry</code>
Research JSON	GET	<code>/research/export</code>

Frontend

Separate app under `frontend/` (`npm install`, `npm run dev`). Not required to verify backend plugin behavior.

Status and limits — 04_Project_Status_and_Roadmap.md